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Information

information 



Easy Forensics picoCTF 2021

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Description

Files can always be changed in a secret way. Can you find the flag?

[cat.jpg](#)

Hints 

1 2

Make sure to submit the flag as
picoCTF{XXXXX}

143,333 users solved



50% Liked



 picoCTF{FLAG}

Submit Flag

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Challenge: Information

Category: Forensics

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I. Objective

The objective of this challenge is to learn how to **analyze file metadata** and identify hidden information within it.

The goal is to extract hidden data from an image file and decode it to **retrieve the flag**.

II. Background

In digital forensics, files often contain **metadata**, which is additional information stored inside the file but not immediately visible.

For image files (like .jpg), metadata can include:

- Camera details
- Author information
- Timestamps
- Custom text fields (like license or comments)

Sometimes, hidden data such as **encoded strings** can be stored in these metadata fields.

In this challenge:

- A string was hidden inside the **metadata of an image**
- The string was encoded using **Base64**
- Decoding it reveals the flag

III. Tool Used

- **Kali Linux (VirtualBox) - Working environment**
- **exiftool - Used to extract metadata from the image**
- **CyberChef - Used to decode Base64 data**

IV. Methodology

1. Launched Kali Linux



2. Downloaded the image file using:

wget <link_address>

```
(kali@kali)-[~/Downloads]
$ wget https://challenge-files.picoctf.net/c_wily_courier/76e95e3e6ee69b4f82b3cea25051f5a9a59
--2026-03-24 04:53:17-- https://challenge-files.picoctf.net/c_wily_courier/76e95e3e6ee69b4f82b
Resolving challenge-files.picoctf.net (challenge-files.picoctf.net)... 3.165.63.36, 3.165.63.56
Connecting to challenge-files.picoctf.net (challenge-files.picoctf.net)|3.165.63.36|:443... con
HTTP request sent, awaiting response... 200 OK
Length: 878136 (858K) [application/octet-stream]
Saving to: 'cat.jpg'

cat.jpg                               100%[=====]
2026-03-24 04:53:21 (638 KB/s) - 'cat.jpg' saved [878136/878136]
```

3. Extracted metadata using:

exiftool cat.jpg

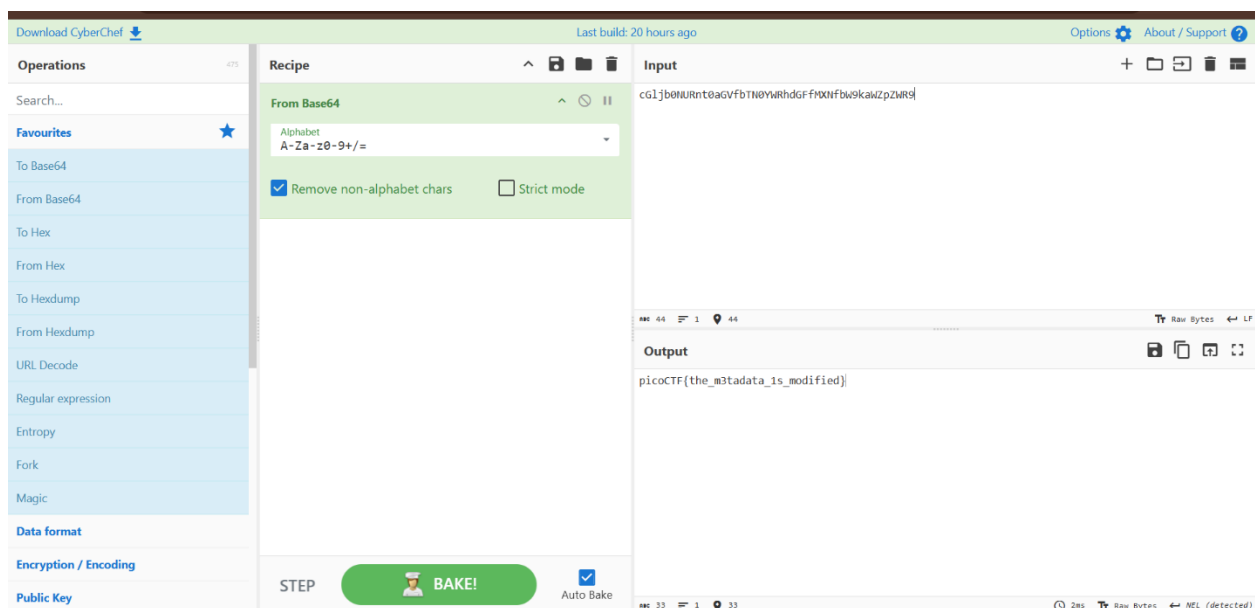
```
(kali@kali)-[~/Downloads]
$ exiftool cat.jpg
ExifTool Version Number      : 13.25
File Name                    : cat.jpg
Directory                    : .
File Size                    : 878 kB
File Modification Date/Time  : 2025:12:12 14:21:14-05:00
File Access Date/Time       : 2026:03:24 04:53:21-04:00
File Inode Change Date/Time  : 2026:03:24 04:53:21-04:00
File Permissions             : -rw-rw-r--
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
JFIF Version                 : 1.02
Resolution Unit              : None
X Resolution                  : 1
Y Resolution                  : 1
Current IPTC Digest          : 7a78f3d9cfb1ce42ab5a3aa30573d617
Copyright Notice             : PicoCTF
Application Record Version   : 4
XMP Toolkit                  : Image::ExifTool 10.80
License                      : cG1jb0NURnt0aGVfbTN0YWRhdGFfMXNfbW
Rights                      : PicoCTF
Image Width                  : 2560
Image Height                 : 1598
Encoding Process              : Baseline DCT, Huffman coding
Bits Per Sample              : 8
Color Components              : 3
Y Cb Cr Sub Sampling         : YCbCr4:2:0 (2 2)
Image Size                   : 2560x1598
Megapixels                   : 4.1
```



4. Observed various metadata fields and identified a suspicious string in the **License** field

```
XMP Toolkit : Image::ExifTool 10.80
License : cGljb0NURnt0aGVfbTN0YWRhdGFfMXNfbW9kaWZpZW99
Rights : PicoCTF
```

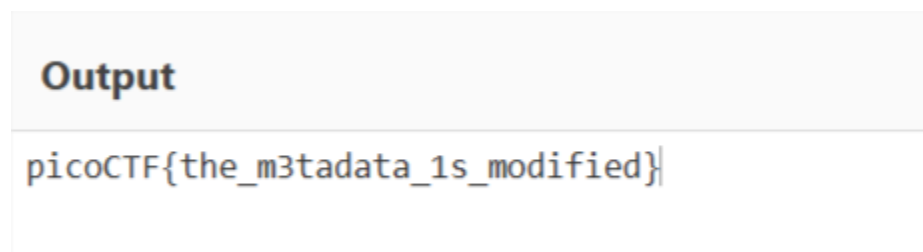
5. Copied the encoded string
6. Opened **CyberChef** and used the **Base64 Decode** function



7. Pasted the string and decoded it
8. The decoded output revealed the **flag**

V. Result

picoCTF{the_m3tadata_1s_modified}





VI. Explanation

What is Metadata?

Metadata is:

“Information about a file, not the file itself”

Example (real-life):

For a photo:

- Image = the picture
- Metadata = details like:
 - when it was taken
 - what camera was used
 - extra hidden notes

What is exiftool?

exiftool is a tool used to:

View hidden information inside files

Command used:

exiftool cat.jpg

It displays all metadata stored in the image.

What did we find?

Inside the metadata, we found something like:

- A strange **random-looking string**
- Located in the **License field**

This is suspicious because:

- It doesn't look like normal text
- It looks encoded

What is Base64?

Base64 is a way of encoding data into text.

It looks like:

aWxpa2V5b3VTYXJhaEc=

How did we decode it?



Using **CyberChef**:

1. Paste the encoded string
2. Apply **Base64 Decode**
3. Output becomes readable text

VII. Conclusion

This challenge demonstrates the importance of analyzing **file metadata in digital forensics**.

By using tools like exiftool, hidden information within the image was discovered. Decoding the Base64-encoded string revealed the flag successfully.

This reinforces a key concept in cybersecurity:

Always check beyond what is visible — valuable data is often hidden beneath the surface.

— **The Analyst: Hyposelenia**